

Dimples Aira Cordon

Laguna State Polytechnic University

Ai.ra Company

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Nature of Business

Custom Software Development, Mobile Application Engineering, and SaaS Solutions Provider

Vision Statement

To be a premier software engineering hub in Southeast Asia, empowering businesses through secure, scalable, and innovative digital solutions.

Mission Statement

[Ai.ra](#) Company delivers enterprise-grade software products by fostering collaborative engineering practices, maintaining strict operational security, and leveraging modern IT infrastructure.

Organizational Structure & Employee Distribution

Organizational Hierarchy:

- **Chief Executive Officer (CEO)**
 - **IT Department Manager**
 - Lead Software Architect
 - Senior Full-Stack Developers (2)
 - **Junior System Administrator**
 - **Finance & Administration Lead**
 - Finance Specialists (4)

- **HR Operations Lead**
 - HR & Recruitment Specialists (3)
- **Sales & Marketing Director**
 - Enterprise Account Executives (5)

PART 2: Enterprise Hardware Inventory

Asset ID	Hardware Category	Item Description	Qty	Department Assignment	Primary Purpose / Justification
HW-DE SK-01	Desktop Workstation	Dell OptiPlex 7010	5	IT	Heavy compiling, virtualization, and local testing environments.
HW-DE SK-02	Desktop Workstation	Dell OptiPlex 5010	5	Finance	Financial modeling, dual-monitor data processing, data reliability.
HW-LAP -01	Laptop Workstation	Lenovo ThinkPad E14	4	HR	On-site interviews, confidential HR record handling, mobility.
HW-LAP -02	Laptop Workstation	Lenovo ThinkPad E14	6	Sales	Client presentations, remote work capabilities, field travel.
HW-SR V-01	Enterprise Rack Server	Dell PowerEdge R450	1	Shared / Server Room	Runs Active Directory, DNS, DHCP, file storage, and local Docker registries.
HW-RT R-01	Enterprise Router	MikroTik Cloud Router Switch CCR2004-16 G-2S+	1	Shared / Network Rack	Handles WAN routing, inter-VLAN routing, and traffic management.

HW-SW-01	Managed PoE+ Switch	Cisco Catalyst 1000 48-Port GbE PoE+ (370W)	1	Shared / Network Rack	Core connectivity for workstations, server, access points, and VoIP.
HW-WAP-01	Wireless Access Point	Ubiquiti UniFi AP AC Pro	2	Shared / Office Floor	Delivers dual-band Wi-Fi connectivity for laptops and mobile devices.
HW-NAS-01	Network Attached Storage	Synology DiskStation DS923+	1	Shared / Server Room	High-capacity automated daily local backup storage.
HW-EXT-01	External Backup Drive	Western Digital Elements 5TB Rugged Portable HDD	2	IT / System Admin	Off-site physical cold backups rotated weekly.
HW-MON-01	Display Monitors	Dell 24" Professional IPS FHD Monitors	20	IT (10), Finance (10)	Dual-monitor configuration for productivity.
HW-PRN-01	Multi-Function Printer	HP LaserJet Enterprise MFP M428fdw	1	Shared / Common Area	Secure printing, scanning, and document handling.
HW-UPS-01	Online Rackmount UPS	APC Smart-UPS SRT 3000VA 230V	1	Shared / Server Room	Uninterruptible power supply protecting network rack and server.
HW-UPS-02	Desktop UPS Units	APC Back-UPS 650VA	10	IT & Finance Desktops	Provides 10 minutes runtime to safely save work during power cuts.

PART 3: Enterprise Software Inventory

Software Name	Version	License Type	Department	Primary Purpose & Selection Justification
Windows 11 Pro	23H2	Commercial OEM / Volume	All	Standard OS providing BitLocker encryption and domain join capabilities for Active Directory management.
Ubuntu Server	24.04 LTS	Open Source (GPL)	Server Room	Enterprise-grade server OS for virtual machines, application hosting, DNS, and internal microservices.
Microsoft 365 Business Standard	Latest Cloud	Commercial Subscription	All	Office tools (Word, Excel, PowerPoint), business email via Exchange, Teams for internal communication, and OneDrive.
Visual Studio Code	1.88+	Open Source (MIT)	IT	Standard lightweight IDE for software development, scripting, and infrastructure management.
Git	2.44+	Open Source (GPL v2)	IT	Distributed version control system for source code tracking.

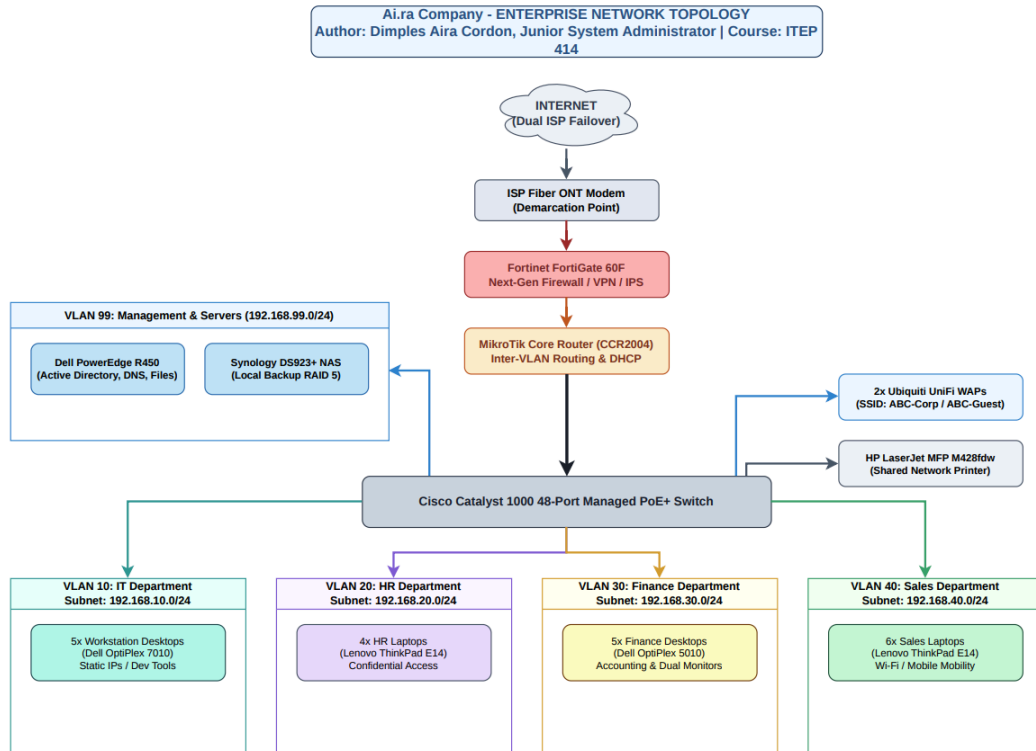
GitHub Desktop	Latest	Open Source (MIT)	IT	Visual GUI interface for developers to manage GitHub repositories efficiently.
Oracle VM VirtualBox	7.0+	Open Source (GPL v3)	IT	Local sandbox virtualization tool for testing isolated Linux deployment environments.
Google Chrome	Latest	Freeware	All	Standard secure web browser across all endpoints with centralized Group Policy management.
Microsoft Defender for Endpoint	Enterprise	Commercial Subscription	All	Enterprise endpoint protection, automated threat detection, and anti-malware solution.
AnyDesk Enterprise	8.0+	Commercial Subscription	IT / Helpdesk	Secure, encrypted remote desktop management utility for internal technical support.
7-Zip	23.01	Open Source (GNU LGPL)	All	File compression and archive extraction tool across workstations.

PART 4: Enterprise Network Inventory

Equipment Name	Quantity	Category	Model / Specification	Functional Role in Network
ISP Modem	1	WAN Gateway	Fiber ONT Modem (Provided by ISP)	Demarcation point converting fiber signals to Ethernet for primary internet uplink.
Enterprise Router	1	Edge Routing	MikroTik CCR2004-16G-2S+	Directs internet traffic, manages subnets, and routes inter-VLAN traffic.
Next-Gen Firewall	1	Security Gateway	Fortinet FortiGate 60F	Performs deep packet inspection, intrusion prevention (IPS), and client VPN hosting.
Managed Core Switch	1	Distribution Switch	Cisco Catalyst 1000 48-Port PoE+	Connects all wired devices, handles VLAN tagging, and powers WAPs via PoE.
Wireless Access Points	2	Wireless Network	Ubiquiti UniFi AP AC Pro	Delivers enterprise Wi-Fi (WPA3 Enterprise) mapped to specific isolated VLANs.

Patch Panel	2	Cabling Infra	24-Port Cat6 Unshielded Patch Panel	Consolidates structured cabling runs from office wall plates into the server rack.
Cat6 Ethernet Cables	1 Box (305m)	Wiring	AMP Netconnect Cat6 UTP Cable	High-speed copper cabling supporting 1Gbps connectivity across the floor.
RJ45 Connectors	1 Box (100pcs)	Connectors	Cat6 Shielded Modular Plugs	Crimp connectors used for custom-length patch cables and wall terminations.

PART 5: Enterprise Network Diagram Description



PART 6: System Administration Roles

1. Helpdesk Technician

- Responsibilities:** Serves as the first point of contact (Tier 1 Support) for end-user technical issues, hardware troubleshooting, operating system re-imaging, basic user account password resets, printer deployment, and ticket tracking.
- Key Skills:** Active listening, operating system troubleshooting (Windows/macOS), hardware diagnostics, basic networking knowledge, and ticketing system workflows.
- Common Tools:** Jira Service Desk, Zendesk, AnyDesk, Sysinternals Suite, Active Directory Users and Computers (ADUC).

- **Certifications:** CompTIA A+, Microsoft Certified: Modern Desktop Administrator Associate.

2. Network Administrator

- **Responsibilities:** Configures, maintains, and secures the physical and logical network infrastructure. Manages routers, switches, firewalls, VPN access, VLAN configurations, bandwidth allocation, and network uptime monitoring.
- **Key Skills:** Routing protocols (OSPF, BGP), VLAN switching, subnetting, network packet analysis, firewall rule management, and VPN setup.
- **Common Tools:** Cisco IOS, Wireshark, SolarWinds Network Performance Monitor, PuTTY/OpenSSH, FortiOS.
- **Certifications:** Cisco Certified Network Associate (CCNA), CompTIA Network+.

3. Linux System Administrator

- **Responsibilities:** Deploys, configures, and maintains Linux server environments. Handles shell scripting, user management, package installation, security patching, system performance monitoring, server backups, and storage management.
- **Key Skills:** Bash scripting, systemd, user/permission management (POSIX/sudo), SSH security, firewall management ([ufw](#)/[iptables](#)), and performance tuning.
- **Common Tools:** Bash, HTOP, Ansible, Cron, Vim/Nano, SSH, LVM (Logical Volume Management).
- **Certifications:** Red Hat Certified System Administrator (RHCSA), CompTIA Linux+, Linux Professional Institute LPIC-1.

4. Cloud Administrator

- **Responsibilities:** Manages enterprise cloud resources across infrastructure-as-a-service (IaaS) and software-as-a-service (SaaS) platforms. Oversees cloud provisioning, IAM policies, cloud security, automated backup storage, cost optimization, and hybrid environment integration.
- **Key Skills:** Identity and Access Management (IAM), cloud networking (VPCs, Subnets), Infrastructure as Code (IaC), cost management, and cloud storage management.

- **Common Tools:** AWS Management Console, Microsoft Azure Portal, Terraform, AWS CLI, Docker.
- **Certifications:** AWS Certified SysOps Administrator, Microsoft Certified: Azure Administrator Associate (AZ-104).

Synergy Across Roles

These four roles function as an interdependent unit to maintain enterprise continuity. The Helpdesk Technician addresses direct end-user issues and escalates complex network or server problems to the specialists. The Network Administrator ensures secure, high-speed data transmission channels across the local office and remote tunnels. The Linux System Administrator builds and maintains the core hosting infrastructure and local application servers that transport this data. Simultaneously, the Cloud Administrator extends local operations into cloud environments, ensuring high availability and off-site data backup. Together, they create a resilient, secure, and fully operational enterprise pipeline.

PART 7: Infrastructure Recommendations

1. Internet Provider (ISP)

- **Recommendation:** Dual ISP Setup — Primary: PLDT Enterprise Fiber (500 Mbps Symmetric Dedicated Line); Secondary: Globe Business Fiber (200 Mbps Backup Line).
- **Justification:** Software development relies heavily on cloud repositories, continuous integration tools, and VoIP/video calls. Implementing dual ISPs with auto-failover at the firewall level ensures zero downtime during line cuts or maintenance.

2. Server Specifications

- **Recommendation:** Dell PowerEdge R450 Rack Server equipped with 2x Intel Xeon Silver 4310 CPUs, 64GB DDR4 ECC RAM, hardware RAID controller running RAID 10 over 4x 2TB Enterprise SAS Drives, and dual redundant power supplies.

- **Justification:** RAID 10 offers the ideal balance of read/write speed for database interactions alongside drive fault tolerance. ECC RAM prevents data corruption, and redundant power supplies protect against internal hardware failures.

3. Backup Strategy (3-2-1 Rule)

- **Recommendation:** Implement the standard 3-2-1 backup strategy:
 - **3 Copies of Data:** Primary production data on the main server.
 - **2 Different Media Types:** Daily automated local snapshots stored on the Synology NAS (Network 1) and local external hard drives.
 - **1 Off-Site Copy:** Encrypted nightly cloud storage backups uploaded to AWS S3 Glacier.
- **Justification:** Guarantees total data recovery capability against localized ransomware attacks, physical hardware failure, or office disaster.

4. Security Recommendations

- **Antivirus / Endpoint Protection:** Centralized deployment of Microsoft Defender for Endpoint managed through Microsoft Intune. Provides real-time behavioral analysis, automated isolation of infected devices, and centralized telemetry.
- **Password Policy:** Enforce domain-level Active Directory policies requiring:
 - Minimum password length of 14 characters.
 - Multi-Factor Authentication (MFA) enforcement across all Microsoft 365 and VPN logins.
 - Mandated password manager usage (Bitwarden) to prevent credential reuse.
 - Account lockout after 5 consecutive failed login attempts.

5. Expansion Plan

- **Recommendation:** The core network switch is intentionally oversized (48 ports for a 20-person team) to accommodate immediate desktop expansion. The network architecture uses /24 CIDR blocks per VLAN, giving each department capacity for up to 254 endpoints without re-subnetting. Additional virtual machines can be spun up on the server by upgrading RAM from 64GB to 128GB.

PART 8: Personal Reflection

Completing this enterprise IT infrastructure planning project provided direct insight into the foundational decisions required before any technical hardware is purchased or deployed. Designing an entire organization's IT environment from scratch highlighted that system administration extends far beyond command-line configurations and troubleshooting; it requires an alignment between technical choices and business operations.

Through this activity, I gained a comprehensive understanding of how hardware allocation, license distribution, and network design directly affect daily productivity and data security. Mapping out employee requirements across four distinct departments reinforced the necessity of matching system specifications to specific workloads—such as assigning high-RAM workstations to developers while providing portable, battery-efficient laptops to field sales staff.

The most challenging task during this project was designing the logical network topology and structuring the inventory subnets. Creating functional VLANs while maintaining proper security boundaries between departments required careful planning. I had to balance security restrictions—such as isolating HR and Finance traffic from general network access—with necessary access to shared resources like server storage and printers. Deciding on the placement of perimeter security devices, such as positioning the Fortinet Firewall between the core router and ISP gateway, deepened my practical understanding of enterprise security design.

This project clearly demonstrated why thorough documentation and planning are critical prior to deployment. Attempting to purchase hardware or deploy networks without a structured inventory leads to resource misallocation, budget overruns, security vulnerabilities, and network failure. Proper documentation serves as a blueprint that reduces downtime, streamlines future system expansions, and allows technical teams to onboard new members efficiently.

Overall, this portfolio project helped me develop a systems-thinking mindset essential for an enterprise administrator. Moving forward, I will apply these skills to design secure,

well-documented, and scalable infrastructure solutions throughout my career in System Administration and Maintenance.